

REPORT CLASSIFICATION

Classification Type	Full Form Analysis
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DATE / TIME

Date of Occurrence	2011-11
Local Time Of Day	1801-2400

PLACE

Locale Reference	Airport
Locale	ZZZ.Airport
State	US
Altitude - AGL	0

ENVIRONMENT

Light	Night
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AIRCRAFT / EQUIPMENT X

Aircraft Operator	Air Carrier
Make Model Name	B737-800
Crew Size	2
Operating Under FAR Part	Part 121
Flight Plan	IFR
Mission	Passenger
Flight Phase	Parked
Maintenance Status - Released For Service	Y
Maintenance Status - Maintenance Type	Unscheduled Maintenance
Maintenance Status - Maintenance Items Involved	Inspection
Maintenance Status - Maintenance Items Involved	Repair

COMPONENT 1

Aircraft Reference	X
Aircraft Component	Fuselage Attachment

Manufacturer	Boeing
Problem	Malfunctioning

COMPONENT 2

Aircraft Reference	X
Aircraft Component	Main Gear
Manufacturer	Boeing

PERSON 1

Location Of Person	Gate / Ramp / Line
Reporter Organization	Air Carrier
Function - Flight Crew	Pilot Flying
Qualification - Flight Crew	Air Transport Pilot (ATP)
ASRS Report Number	978777
Human Factors	Communication Breakdown
Human Factors	Confusion
Communication Breakdown - Party1	Flight Crew
Communication Breakdown - Party2	Maintenance
Communication Breakdown - Party2	Other
Callback	Completed

EVENTS

Anomaly	Aircraft Equipment Problem - Less Severe
Anomaly	Deviation / Discrepancy - Procedural - FAR
Anomaly	Deviation / Discrepancy - Procedural - Published Material / Policy
Detector - Person	Flight Crew
Were Passengers Involved In Event	No
When Detected	Pre-flight
Result - General	Flight Cancelled / Delayed
Result - General	Maintenance Action

ASSESSMENTS

Contributing Factors / Situations	Aircraft
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Contributing Factors / Situations	Company Policy
Contributing Factors / Situations	Human Factors
Contributing Factors / Situations	Procedure
Primary Problem	Aircraft

NARRATIVE 1

Three out of four of my past flights now have had significant delays due the known issue of "wheel well vapor barrier cracking." I have personally found the first evidence of this in two aircraft in our fleet with this cracking issue. It is referred to and repaired In Accordance With (IAW) Boeing Structural Repair Manual (SRM) 57-10-10-1A2. Maintenance Control advises we have now at least 24, B737 aircraft with this cracking issue. I am fully aware of the non-structural pertinence of this issue. I am, though, fully aware that Boeing and all carriers are waiting until cracks are found before tending to them.

Herein lies the issues with rather large flight delays. I strongly suggest and request more proactive involvement by our Maintenance personnel to find these cracks sooner by increased inspections. The current SRM quoted above allows for a crack not to exceed 16 inches. I found last week (first report I might add on this particular aircraft X) a crack 17.5 inches long. Aircraft X was allowed to fly the next day after Engineering issued a Line Authorization. Aircraft X was taken out of service while waiting for this Engineering issuance arrival. The longest crack now allowed has been extended to 17.5 inches, directly due to my discovery of this crack on aircraft X.

This morning, I found Aircraft Y with a crack in the left wheel well (aircraft X was in the right wheel well). Just as in aircraft X, aircraft Y had a smaller crack 7 inches [long] and my discovery was, again, the first report of vapor barrier cracking on aircraft Y. The third, aircraft Z, I found cracking had an approximately 4 inch extended crack beyond a previously repaired section. I found the crack progression on that aircraft last week. I have found that these cracks do not stop progressing until they hit the bottom of the vapor barrier and finally run almost to the top, on a 45-degree angle, running inboard and upward. An interesting thing I saw this morning on aircraft Y, that I highly recommend Boeing Engineers look at, is the approximate 6 x 6 inch open area of first theoretical cracking in the right wheel well, as of yet un-cracked, I noticed that this area was significantly deformed. The center of the area was "oil canned" forward, away from the wheel well. If one would put a straight edge across the area, I estimated the distortion was on the order of approximately 3/16 of an inch. Thereby, indicating a great amount of stress in that area. It is obvious that in very short order, this area will also be cracked as the left wheel was this morning and the exact area was cracked on aircraft X.

Please see the following description. Note: the most displaced part of the crack running from upper left to bottom right. The physical dimension of the crack seen at this point is just over 6 inches. This is viewed from the right wheel well, standing in the center of the wheel well, facing forward. Crack is approximately, directly in front of a 6 foot tall man's right collar bone and the far outboard, lower edge of the crack, is at the bottom of

the vapor barrier panel. There are three vertical rivets on a bulkhead fitting. In between the middle rivet and the next rivet to the right that is in a vertical row of six rivets on a vertical former, the upper edge of the crack can be seen as a 45-degree angle sloping down from upper left to bottom right, again just to the right of the middle left three rivets. This is the far upper left edge of the crack and ends under the bulkhead fitting. Again, this crack was objectively measured by two mechanics to be 17.5 inches long and is now the maximum allowed crack, regarding this issue, by Boeing.

A very descriptive Service Bulletin is greatly needed with inspections made routinely. Our operation is being severely effected. I am always available on my own time to assist in any way I can.

CALLBACK 1

Reporter stated the vapor barriers are flat segmented rubber-like blade seals that are installed around the inside of the left and right main wheel well openings of the lower fuselage body fairing. The fairing is a fiberglass type material. Each time the Main Landing Gear (MLG) is cycled, the MLG tires brush-up and down across the seals causing stress on the seal retainers, which are attached to the fuselage lower fiberglass fairing. The reporter stated he did not believe this to be a safety of flight issue, but rather a FAR and Maintenance compliance issue under their Procedures Manual (PM).

SYNOPSIS

A pilot reports numerous B737 NexGen aircraft are being found with lower fuselage body fairings cracking in the vertical sections that support the left and right main wheel well vapor barrier blade seals. The blade seals are attached to the inside area of the fiberglass fairing.

